

Beetroot (*Beta vulgaris*) – Risk–Benefit Assessment

1. Executive Summary

Beetroot (*Beta vulgaris*) is a root vegetable and functional food rich in nitrates, betalain pigments, fibre, folate, potassium and polyphenols. It is consumed as a whole vegetable, juice, powder and as a natural colourant (E162) in foods.

At typical dietary intakes, beetroot is considered low-risk for healthy adults and is associated with modest reductions in blood pressure, improved exercise performance and potential cardiovascular and anti-inflammatory benefits. Concentrated beetroot juice provides pharmacological doses of nitrate that can lower systolic blood pressure by around 4–8 mmHg in many trials.

Key risks are largely dose- and context-dependent: high oxalate load (kidney stone risk), nitrate load (hypotension and theoretical nitrosamine formation), and high potassium for those with advanced kidney disease. Overall, risk is low for normal food use but becomes moderate in individuals with kidney disease, recurrent stones or on multiple cardiovascular drugs using frequent, high-dose juice or supplements.

2. Definition & Context

- **Identity:** Beetroot is the taproot of *Beta vulgaris* subsp. *vulgaris*, family *Amaranthaceae*. Its key bioactive constituents include inorganic nitrate (NO_3^-), betalain pigments (especially betanin), betaine, polyphenols and dietary fibre.
- **Colourant:** The purified red pigment (betanin) is authorised as the food colour “Beetroot red” or E162 (CAS 7659-95-2). It is widely used in beverages, desserts, dairy products, confectionery and other processed foods.
- **Uses:** Whole beets (cooked/roasted), juices and shots, smoothie blends, powders, capsules and sports/“pre-workout” formulations. Exposure is overwhelmingly via ingestion; skin and inhalation exposure are only relevant for workers handling powders.
- **Regulatory context:** Beetroot itself is treated as a conventional food. Beetroot red (E162) has been re-evaluated by EFSA and is considered not of safety concern at currently reported use levels. Dietary nitrate has an Acceptable Daily Intake (ADI) of 0–3.7 mg/kg body weight (expressed as nitrate ion).

3. Benefits / Positive Effects

Benefit	Why It Matters (real-world impact)	Evidence / Source (example)
Blood pressure reduction in hypertension	Nitrate-rich beetroot juice can lower systolic blood pressure by ~4–8 mmHg in many adults with elevated BP, supporting cardiovascular risk reduction when used alongside standard care.	Multiple RCTs and meta-analyses using beetroot juice providing ~200–800 mg nitrate/day for weeks.
Improved exercise performance and endurance	Enhanced exercise economy and time-to-exhaustion in some athletes and recreationally active adults; practical for endurance events, high-intensity intervals and team sports.	Umbrella reviews of beetroot juice as a nitrate source show modest improvements in VO ₂ efficiency and time-trial performance.
Endothelial and vascular health	Dietary nitrate from beetroot increases nitric oxide bioavailability, improves arterial dilation and may reduce arterial stiffness, supporting long-term vascular health.	Human trials show improved flow-mediated dilation and reduced arterial stiffness after beetroot juice intake.
Anti-inflammatory and antioxidant effects	Betalains and polyphenols scavenge reactive oxygen species and modulate inflammatory pathways, which may help limit chronic low-grade inflammation and oxidative stress.	Cell, animal and small human studies show reductions in inflammatory markers and oxidative damage with beetroot intake.
Metabolic and liver support	Beetroot juice has shown improvements in markers of non-alcoholic fatty liver disease in some small trials, especially when combined with dietary change, and may modestly improve lipid profiles.	Pilot RCTs in NAFLD patients using ~250 mL beetroot juice/day over 8–12 weeks.
Potential cognitive benefits (select settings)	By improving cerebral blood flow, beetroot-derived nitrate may acutely support certain cognitive tasks, though longer-term data are mixed.	Acute trials in healthy adults show improved cerebral perfusion and some cognitive tasks; a 13-week trial in older adults showed no clear benefit.
Natural colourant alternative	Beetroot red (E162) offers a natural alternative to some synthetic red dyes in foods and beverages, with a favourable safety profile at current use levels.	EFSA re-evaluation of E162 concluded no safety concern at reported exposure levels.

4. Risks / Negative Effects

Risk / Effect	Potential Impact (mild / moderate / severe)	Type of Evidence	Notes
Beeturia (red/pink urine and stools) – acute	Mild; benign but can alarm patients.	Human observational and case reports	Pigment excretion after eating beets or juice; not harmful and resolves on stopping intake.
Gastrointestinal upset – acute	Mild-moderate bloating, cramps or diarrhoea in some people.	Human self-reports and small trials	Likely related to FODMAPs, fibre and osmotic load of juice; more common with large volumes or on empty stomach.
Excess blood pressure lowering / dizziness – acute	Mild-moderate; may be severe in frail or over-treated hypertensive patients.	Human trials and clinical experience	Nitrate-induced vasodilation can interact with antihypertensives or PDE-5 inhibitors; monitor BP in those on multiple agents.
Kidney stone promotion (oxalate) – chronic	Moderate in people with a history of calcium-oxalate stones; low in others.	Case reports and mechanistic data	Beetroot is high in oxalate. Very high, regular intakes (especially juices plus other high-oxalate foods) have been implicated in oxalate nephropathy and stone formation.
Potassium load in advanced CKD – chronic	Potentially severe hyperkalaemia risk in advanced kidney failure.	Nutrient composition data + nephrology guidance	Beetroot and juice provide substantial potassium. In advanced CKD or on ACE-i/ARB/spironolactone, high intakes should be avoided without medical supervision.
Nitrate / nitrite and nitrosamine formation – chronic	Uncertain; theoretical cancer risk at high nitrate/nitrite exposure, but vegetable-derived nitrate appears low-risk.	Regulatory and epidemiological assessments	WHO/EFSA set an ADI of 0–3.7 mg/kg bw for nitrate. EFSA concludes that nitrate from vegetables is unlikely to increase cancer risk and benefits of vegetable intake dominate.
Allergic reactions – acute	Rare; can be mild to severe.	Isolated case reports	As with most plant foods, IgE-mediated allergy is possible but uncommon; manifests as oral itching, rash or, rarely, anaphylaxis.
Colourant exposure (E162)	Low concern at current use levels.	Regulatory evaluations (EFSA)	No numerical ADI considered necessary; no evidence of genotoxicity or carcinogenicity at typical exposures.

5. Toxicity & Safety Profile

Overall toxicity: Beetroot as a food has a long history of safe use. The main toxicological issues relate to its nitrate, oxalate and potassium content at high or concentrated intakes.

Nitrate: The Joint FAO/WHO Expert Committee on Food Additives (JECFA) and EFSA set an ADI for nitrate of 0–3.7 mg/kg body weight per day (as nitrate ion). For a 70 kg adult this is about 260 mg nitrate/day. Beetroot nitrate content is highly variable with growing conditions: values from ~25 mg/100 g to over 100 mg/100 g fresh weight are reported, while concentrated juices can provide 200–800 mg nitrate in a single daily dose used in clinical trials. Short-term trials above the ADI have not revealed serious toxicity in healthy adults, but the long-term safety of chronic high-dose supplementation is less well established.

Nitrite and methaemoglobinaemia: Infants under 3–6 months are especially vulnerable to nitrate/nitrite-induced methaemoglobinaemia (“blue baby syndrome”). For this reason high-nitrate vegetables such as beetroot and spinach are generally not recommended as early weaning foods or in home-made juices for young infants.

Oxalate and kidney: Beetroot is a high-oxalate food. Occasional case reports link heavy consumption of beetroot and other high-oxalate juices with oxalate nephropathy and stone formation, particularly in people with underlying kidney disease or low fluid intake. For stone-prone individuals, moderation and good hydration are important.

Colourant E162: EFSA’s re-evaluation of beetroot red concluded that, based on available toxicological data and the low exposures from use as a food colour, there is no safety concern and no numerical ADI is required. It is not classified as carcinogenic, mutagenic or a reproductive toxicant.

Vulnerable groups:

- Infants <6 months – avoid beetroot juices/purees as primary foods due to nitrate risk.
- People with recurrent calcium-oxalate kidney stones – limit large, frequent servings of beetroot juice; prefer modest portions of whole beetroot.
- Advanced chronic kidney disease or on multiple RAAS-blocking drugs – monitor potassium and avoid daily high-volume juice.
- Individuals with very low baseline blood pressure or on multiple antihypertensives – introduce beetroot juice cautiously and monitor BP.

6. Environmental Impact

Beetroot as a crop behaves similarly to other intensively grown vegetables.

- Nitrate leaching: Beets accumulate nitrate from fertilised soils. If nitrogen fertiliser use is excessive or poorly managed, nitrate can leach into groundwater and run off into surface

waters, contributing to eutrophication and drinking-water contamination.

- Persistence and bioaccumulation: The main bioactives (nitrate, betalains, betaine) are water-soluble and biodegradable; they do not bioaccumulate in wildlife in the classical sense seen with persistent organic pollutants.
- Colourant E162: As a plant-derived pigment, beetroot red is readily degradable in the environment and not known to pose specific ecotoxicological concerns at current usage levels.

From an environmental perspective, the main issue is fertiliser-driven nitrate pollution from intensive vegetable cultivation rather than beetroot consumption itself.

7. Who Should / Should Not Be Exposed

General healthy adults:

- Whole beetroot several times per week is safe and nutritionally beneficial.
- Beetroot juice up to ~250 mL/day is reasonable for most adults, especially those targeting blood-pressure or endurance benefits.

Children and adolescents:

- Small portions of cooked beetroot as part of a varied diet are acceptable.
- Avoid concentrated juices and shots in very young children; focus on whole foods.

Pregnant and breastfeeding women:

- Moderate food-level beetroot intake is acceptable and can contribute folate and other nutrients.
- Avoid aggressive “detox” juicing regimens that deliver very high nitrate and oxalate loads.

Kidney disease / history of stones:

- Prefer modest portions of whole beetroot rather than daily large glasses of juice.
- Discuss regular beetroot juice use with a nephrologist, especially in advanced CKD or recurrent stone formers.

People with low blood pressure or on multiple cardiovascular drugs:

- Introduce beetroot juice cautiously; monitor for dizziness or symptomatic hypotension.
- Time juice away from peak drug effect if used as an adjunct.

Workers handling beetroot powders or concentrates:

- Standard food-handling hygiene and dust control are generally sufficient; serious occupational hazards are not described.

8. Safer Alternatives or Mitigation Strategies

If the goal is cardiovascular or performance benefit:

- Use modest doses: 70–250 mL/day of nitrate-rich beetroot juice rather than very large volumes.
- Prefer products with analysed nitrate content to avoid massive variability.
- Combine with a high-vegetable, high-fruit dietary pattern (e.g. Mediterranean style) rather than relying on juice alone.

To reduce kidney stone risk:

- Prefer whole beets over concentrated juice; keep portions moderate.
- Ensure high fluid intake; avoid combining large beetroot juice volumes with other high-oxalate juices daily.

To reduce nitrate/nitrosamine concern:

- Emphasise fresh vegetables and fruits, adequate vitamin C and polyphenol intake (they inhibit nitrosamine formation).
- Avoid storing beetroot juice for long periods; consume within 24 hours and keep chilled.

For those who should limit beetroot:

- Alternate nitrate sources include other vegetables (e.g. lettuce, rocket, celery) but these also carry nitrate; for high-risk groups, blood-pressure control should prioritise lifestyle and medication rather than heavy nitrate supplementation.

9. Comparison with Alternatives

Option	Pros	Cons	Regulatory / Safety Notes	Best For
Beetroot juice (nitrate-rich)	Standardised, high nitrate dose; good evidence for BP lowering and exercise performance; easy to consume.	High oxalate and potassium load; sugar content; BP may drop too much in susceptible people; large doses may exceed nitrate ADI.	Food, not a drug; no specific upper limit, but general nitrate ADI applies as a guide.	Adjunct for hypertension and endurance performance in otherwise healthy adults under medical supervision.
Beetroot powder / capsules	Convenient, portable, often lower sugar; can be combined with other nutrients.	Nitrate content highly variable between brands; evidence base smaller than liquid juice.	Regulated as food supplement; quality and nitrate content depend on manufacturer.	People wanting convenience or sugar control while still accessing some beetroot benefits.

Inorganic nitrate salts (e.g. sodium/potassium nitrate supplements)	Precise dosing of nitrate; cheap.	Lack the antioxidant and polyphenol matrix of vegetables; higher theoretical risk if overused; less palatable.	More clearly constrained by nitrate ADI; excessive use can breach safe limits quickly.	Niche sports or research settings with expert oversight, not routine consumer use.
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10. Key References (Selection)

- JECFA / WHO – Nitrate and nitrite evaluations; ADI 0–3.7 mg/kg bw for nitrate.
- EFSA re-evaluation of Beetroot red (E162) – no safety concern at reported use levels.
- Systematic reviews and meta-analyses of beetroot juice and blood pressure / exercise performance (various, 2015–2025).
- Chen L. et al., 2021. Beetroot as a functional food with huge health benefits – overview of composition and bioactivity.
- Zamani H. et al., 2021. The benefits and risks of beetroot juice consumption – critical review.
- Recent clinical trials and umbrella reviews on beetroot juice nitrate and physical performance (e.g. Tian C. et al., 2025).